



International Islamic University Chittagong (IIUC)
Department of Computer Science and Engineering (CSE)
Semester Final Examination

Program: B. Sc. in CSE	Semester: Autumn-2024
Course Code: MATH-2407	Course Title: Mathematics-IV
Time: 2:30 hours	Total Marks: 50

- (i) Answer all the questions. The figures in the right-hand margin indicate full marks.
- (ii) Separate answer script must be used for separate group.
- (iii) Course Learning Outcomes (CLOs) and Bloom's Levels are mentioned in additional Columns.

Course Learning Outcomes (CLOs) of the Questions

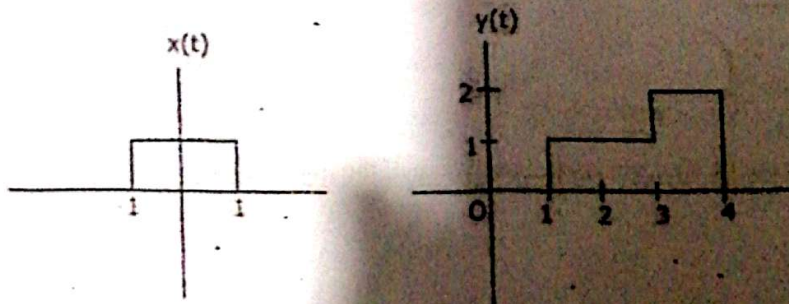
CLO1	Demonstrate the understanding of the basic principles and operations set theory, complex numbers, geometrical interpretation complex functions and the concept of transformation in a complex plane.
CLO2	Apply the concept of transformation of an object into complex space and operation of complex functions
CLO3	Use Fourier series, Laplace's Transforms, Fourier Transform in different scenario.
CLO4	Analyze the harmonics & spectrum of different types of waves.
CLO5	Demonstrate the harmonic analysis using MATLAB.

Bloom's Taxonomy Domain Levels of the Questions

Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Group-A

- | | | | |
|--|---------------------------------------|---|-------------------------------------|
| <p>1. Sketch 3 cycles of the function represented by</p> <p>$y = f(t) = 3t ; \quad 0 < t < 1$
 $\quad \quad \quad = 3 ; \quad 1 < t < 2$
 $f(t+2) = f(t)$ Here, $T = 2L = 2 \quad \therefore L = 1$</p> <p>(ii) Find the Fourier series for the above function.</p> <p>2. Evaluate the convolution sums of $y[n] = x[n] * h[n]$</p> <p>Where,</p> <p style="text-align: center;">$x[n] = \begin{cases} 1, & n = 0 \\ 3, & n = 1 \end{cases}$ and $h[n] = \begin{cases} 2, & n = 0 \\ 1, & n = 1 \end{cases}$</p> <p>Where, n represents the time index.</p> <p>Or</p> <p>a) Find the convolution integral of $x(t) * y(t)$</p> | <p>Mark
2+8</p> <p>6</p> <p>6</p> | <p>CLO
CLO3</p> <p>CLO4</p> <p>CLO4</p> | <p>DL
App</p> <p>U</p> <p>U</p> |
|--|---------------------------------------|---|-------------------------------------|



- b) i. If $x = \cos 4t + \frac{1}{3} \sin 4t$ then find the amplitude of the new signal
 ii. How would you explain the functioning of the cochlea inside the ear in terms of Fourier Analysis?

1 CLO4 An

3 CLO4 An

Or

- 15 Plot the line (at least 4) spectrum for the following complex wave

4 CLO4 An

$$\underbrace{f(t)}_{\text{Complex wave}} = \underbrace{\frac{2\pi^2/3}{2}}_{\text{DC value}} + \sum_{n=1}^{\infty} \underbrace{\frac{4}{n^2} \cos n\pi}_{\text{Amplitude}} \underbrace{\cos nt}_{\text{Frequency}}$$

Group-B

3. a) Find the Fourier transform of the function,

Mark 5 CLO3 DL Ap

$$f(t) = \begin{cases} 1, & 0 < t < 1 \\ -1, & -1 < t < 0 \\ 0, & \text{otherwise} \end{cases}$$

Or

- a) Find the Laplace transform of the functions $f(t) = \cos at$

5 CLO3 Ap

- b) Express the following function in terms of unit step functions and find its Laplace transform:

5 CLO3 Ap

$$f(t) = \begin{cases} 10; & t \leq 3 \\ 8; & t \geq 3 \end{cases}$$

Or

- b) Solve the following Initial Value Problem (IVP) by Laplace Transform

5 CLO3 Ap

$$Y'' + Y = t \quad Y(0) = 1 \quad Y'(0) = -2$$

4. a) Evaluate $L\{e^{-3t}\}$ using First Shift theorem.

2 CLO3 Ap

- b) Why do the lines of an aperiodic signal look so close together when plotted as a spectrum?

2 CLO3 U

- c) Draw the Function

6 CLO3 U

$$f(t) = -u(t+3) + 2u(t+1) - 2u(t-1) + u(t-3)$$

5. a) Write down a user defined function in MATLAB to reconstruct $f(t)$ in the time interval of $[-4, 20]$ for the following complex wave and also draw line Spectrum for the following wave in one window

8 CLO5 An

$$f(t) = 2\pi + \sum_{n=1}^{500} \frac{4}{n} \sin n\pi$$

- b) If

2 CLO5 An

$$x[n] = 2; n = 0$$

$$h[n] = 2; n = 0$$

$$= 6; n = 1$$

and

$$= -2; n = 1$$

Write MATLAB code to find the convolution sum of the above signals.



International Islamic University Chittagong (IIUC)
Department of Computer Science and Engineering (CSE)
Semester Final Examination

Program: B. Sc. in CSE
Course Code: MATH-2407
Time: 2:30 hours

Semester: Spring-2024
Course Title: Mathematics-IV
Total Marks: 50

- (i) Answer all the questions. The figures in the right-hand margin indicate full marks.
(ii) Please answer the several parts of a question sequentially.
(iii) Separate answer script must be used for separate group.
(iv) Course Learning Outcomes (CLOs) and Bloom's Levels are mentioned in additional Columns.

Course Learning Outcomes (CLOs) of the Questions

CLO1:	Demonstrate the understanding of the basic principles and operations set theory, mathematical operations with complex numbers, geometrical interpretation and the related fundamental theories involving complex functions and the concept of transformation in a complex plane.
CLO2	Understanding the behavior of complex wave, periodic functions, and aperiodic functions, stable signal, unstable signal, impulse signal, unit step function, ramp function, convolution theorem.
CLO3:	Use Fourier series, Laplace's Transforms, Inverse Laplace's Transform, Fourier Transform in different scenario.
CLO4:	Analyze the harmonics & spectrum of different types of waves.
CLO5:	Demonstrate the harmonic analysis using MATLAB.

Bloom's Taxonomy Domain Levels of the Questions

Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

GROUP-A

- | | | | | | |
|----|----|---|-------|-------------|-----------|
| 1. | a) | What is obtained by breaking down a complex wave? | Marks | CLO | DL |
| | b) | Find the complex form of the periodic function $f(t)$ is given by: | 01 | CLO2 | U |
| | | $f(t) = \cos \frac{1}{2}t \quad ; -\pi < t < \pi \quad [T = 2L = 2\pi]$ | 09 | CLO3 | U |

- 2 a) Evaluate the convolution sums of $y[n] = x[n] * h[n]$
Where,

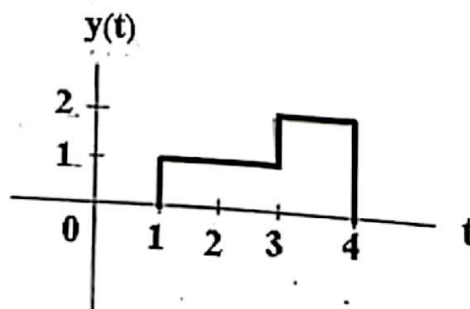
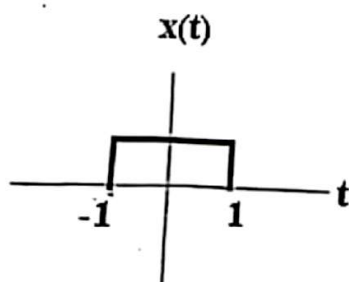
$$x[n] = \begin{cases} 1, & n = 0 \\ 3, & n = 1 \end{cases} \text{ and } h[n] = \begin{cases} 2, & n = 0 \\ 1, & n = 1 \end{cases}$$

08 **CLO2** **U**

Where, n represents the time index.

Or

Find convolution integral of $x_1(t) * x_2(t)$



- b) If $x = \cos 3t + \frac{1}{4} \sin 4t$ then find the amplitude of the new signal and the phase shift.

2 CLO2 U

Or

$$y = f(t) = 0; \quad -\pi \leq t < 0$$

$$= t; \quad 0 \leq t < \pi$$

Here, $T = 2L = 2\pi$

$$\therefore L = \pi$$

Sketch the function for 3 cycles

GROUP-B

- | | | Marks | CLO | DL |
|----|--|-------|------|-----|
| 3. | a) Find the Fourier transform of $f(t) = \begin{cases} 1 + \frac{t}{a}, & -a < t < 0 \\ 1 - \frac{t}{a}, & 0 < t < a \\ 0, & \text{otherwise} \end{cases}$ | 5 | CLO3 | U |
| | Or Find the Laplace transform of the functions: $f(t) = \sin at$ | | | |
| | b) Express the following function in terms of unit step functions and hence find its Laplace transform $f(t) = \begin{cases} 8; & t < 2 \\ 6; & t > 2 \end{cases}$ | 5 | CLO3 | U |
| 4. | a) Evaluate $\mathcal{L}\left\{\frac{\sin 2t}{t}\right\}$ using the division theorem. | 5 | CLO3 | U |
| | Or Solve the Initial Value Problem (IVP) by Laplace Transformation, $Y'' - 3Y' + 2Y = 4e^{2t}$, $Y(0) = -3$, $Y'(0) = 5$ | | | |
| | b) Draw the graph of the following functions
i. $-2u(t+3)$ ii. $x(t-2)$ iii. $3\delta(t-2)$ | 3 | CLO2 | U |
| | c) Plot the line spectrum (at least 2) for the following Fourier series:
$f(t) = \frac{\pi^2}{3} + \sum_{n=1}^{500} \frac{(-1)^{2n}}{n^2} \cos n\pi t$ | 2 | CLO2 | U |
| 5. | a) Write a user defined MATLAB function to reconstruct $f(t)$ in the time interval of $[-4, 20]$ for the following complex wave
$f(t) = 2\pi + \sum_{n=1}^{500} \frac{2}{n} \cos 2n\pi t$ | 4 | CLO5 | App |
| | b) Write MATLAB code for the following unit step function
$5u(t) + 6u(t+2) - 7u(t-1)$ | 4 | CLO5 | App |
| | c) Write MATLAB code to find the convolution sums of $y[n] = x[n] * h[n]$
Where,
$x[n] = \begin{cases} 2, & n = 0 \\ 3, & n = 1 \end{cases} \text{ and } h[n] = \begin{cases} 3, & n = 0 \\ 1, & n = 1 \end{cases}$
Where, n represents the time index. | 2 | CLO5 | App |



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Semester Final Examination

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Time: 2:30 hours	Total Marks: 50

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Course Learning Outcomes (CLOs) of the Questions

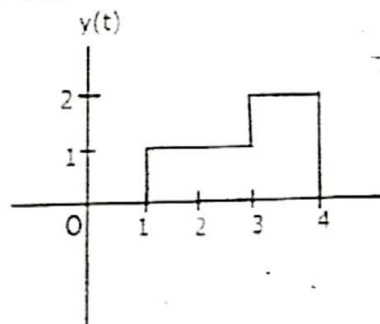
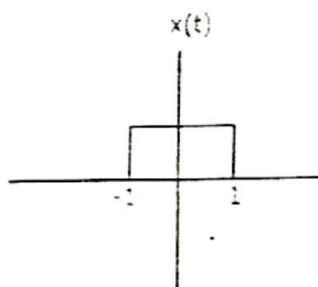
CLO1	Demonstrate the understanding of the basic principles and operations set theory, complex numbers, geometrical interpretation complex functions and the concept of transformation in a complex plane.
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CLO5	Demonstrate the harmonic analysis using MATLAB.

Bloom's Taxonomy Domain Levels of the Questions

Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Group-A

- | | Mark | CLO | DL |
|---|-------------|------------|-----------|
| 1. If $y = f(t) = t^2$ over the interval $-\pi < t < \pi$ and has period 2π . Here,
$T = 2L = 2\pi \therefore L = \pi$ | 5 | CLO3 | U |
| a) Find the Fourier series for the above function. | 3 | CLO3 | U |
| b) Hence, deduce $\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots = \frac{\pi^2}{12}$ | 2 | CLO1 | R |
| c) Write down the complex form of Fourier series. | 6 | CLO4 | U |
| 2. a) Evaluate the convolution sums of $y[n] = x[n] * h[n]$
Where,
$x[n] = \begin{cases} 1, & n = 0 \\ 3, & n = 1 \end{cases} \text{ and } h[n] = \begin{cases} 2, & n = 0 \\ 1, & n = 1 \end{cases}$
Where, n represents the time index.
Or
Find the convolution integral of $x(t) * y(t)$ | | | |



- b) i. If $x = \cos 4t + \frac{1}{3} \sin 4t$ then find the amplitude of the new signal and the phase shift. 2+2=4 CLO4 An
 ii. Is there any difference in line spectrum in Fourier series and Fourier transform? What kind of difference? And why?
 Or
 Plot the line (at least 4) spectrum for the following complex wave 4 CLO4 An

$$f(t) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{1}{2n-1} \cos nt + \sum_{n=1}^{\infty} \frac{(-1)^n}{2n} \sin nt$$

Group-B

3. a) Find the Fourier transform of the function, Mark 5 CLO CLO3 DL Ap

$$f(t) = \begin{cases} 1, & 0 < t < 1 \\ -1, & -1 < t < 0 \\ 0, & \text{otherwise} \end{cases}$$

 Or
 Find the Laplace transform of the functions $f(t) = \cos at$
- b) Express the following function in terms of unit step functions and find its Laplace transform: 5 CLO3 Ap

$$f(t) = \begin{cases} 10; & t \leq 3 \\ 8; & t \geq 3 \end{cases}$$

 Or
 Solve the following Initial Value Problem (IVP) by Laplace Transform
 $Y'' + 4Y = 12t \quad Y(0) = 0 \quad Y'(0) = 7$
4. a) Evaluate $\mathcal{L}\left\{\frac{\sin 2t}{t}\right\}$ using the division theorem. 5 CLO3 Ap
 b) Draw the Impulse Function $2 * \delta(t-3)$ 1 CLO3 U
 c) Draw the Function $x(t) = r(t+1) - r(t) + r(t-2)$ 4 CLO3 U
5. a) Write down a user defined function in MATLAB to reconstruct $f(t)$ in the time interval of $[-4, 20]$ for the following complex wave 5 CLO5 An

$$f(t) = 10 + \frac{10}{3\pi} \left(\sin \frac{\pi t}{4} + \frac{1}{3} \sin \frac{3\pi t}{4} + \frac{1}{5} \sin \frac{5\pi t}{4} + \frac{1}{7} \sin \frac{7\pi t}{4} + \dots \right)$$

 b) Write down MATLAB code for the following function: 3 CLO5 An

$$x(t) = -u(t+3) + 2u(t+1) - 2u(t-1) + u(t-4)$$

 c) If 2 CLO5 An

$$x[n] = 2; n=0 \quad \text{and} \quad h[n] = 2; n=0$$

$$= 6; n=1 \quad \quad \quad = -2; n=1$$

 Write MATLAB code to find the convolution sum of the above signals.

International Islamic University Chittagong (IIUC)
Department of Computer Science and Engineering (CSE)
B. Sc. in CSE; Semester Final Examination; Spring-2023

Course Code: MATH-2407

Course Title: Mathematics-IV

Time: 2:30 hours

Total Marks: 50

Answer all the questions. The figures in the right-hand margin indicate full marks.

Group A

1. Define Fourier series in the interval $(-L, L)$. Sketch the following function for four cycles, also find the Fourier series for the function. 10 CO3 U

$$y = f(t) = \begin{cases} 0; & -3 \leq t < 0 \\ 3; & 0 \leq t < 3 \end{cases}$$

- Or Derive the complex form of the Fourier series. 10 CO3 U

- 2.a Find Harmonic analysis of the given Fourier series 5 CO4 An

$$f(t) = \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n+2}}{n} \sin \frac{n\pi t}{6}$$

- 2.b Find the Convolution Sum $x[n] * h[n]$ for the following functions 5 CO4 An

$$\begin{aligned} x[n] &= 3; \quad n=0 \\ &= 1; \quad n=1 \end{aligned} \quad \text{and} \quad \begin{aligned} h[n] &= 1; \quad n=0 \\ &= 3; \quad n=1 \end{aligned}$$

Where n is the time index

Group B

- 3.a Solve the following Initial Value Problem (IVP) by Laplace Transform: 5 CO3 Ap
 $Y'' + Y = t, \quad Y(0) = 1, \quad Y'(0) = 2$

- Or Find Laplace Transform $t^2 \sin 2t$ using the Multiplication theorem. 5 CO3 Ap

- 3b Sketch the waveforms for the following signals 5 CO1 Ap
 $x(t) = -u(t+3) + 2u(t+1) - 2u(t-1) + u(t-3)$

- Or $x(t) = r(t+2) - r(t+1) - r(t-1) + r(t-2)$ 5 CO1 Ap

- 4.a State first shift theorem. Using the theorem evaluate $\mathcal{L}\{e^{-4t}t^2\}$ 5 CO1 U

- 4.b Define unit step function. Sketch the following function, 5 CO1 U
 $x(t) = -u(t+3) + 2u(t+1) - 2u(t-1) + u(t-3)$

- 5.a Write MATLAB code to sketch line spectrum (at least 6) for the following 4 CO5 Ap

$$\text{Fourier series } \underbrace{f(t)}_{\text{Complex wave}} = \underbrace{2}_{\text{DC value}} + \underbrace{\left[\sum_{n=1}^{\infty} (\cos 2n\pi + 1) \sin \frac{n\pi t}{6} \right]}_{\text{AC value}}$$

- 5.b Make a function in MATLAB environment to raise a complex wave $f(t)$ in the time interval of $[-4, 20]$ for the following Fourier series: 6 CO5 Ap

$$f(t) = 4\pi + \sum_{n=1}^{\infty} \frac{4}{n\pi} \cos 2n\pi t$$

International Islamic University Chittagong
Department of Computer Science & Engineering

B.Sc. in CSE Final Examination, Autumn 2022

Course Title: Mathematics-IV Course Code: MATH-2407 (New)

Course Title: Mathematics-V Course Code: MATH-3501 (Old)

Time: 2 Hours 30 Minutes

Full Marks: 50

[The figures in the right-hand margin indicate full marks, Separate answer scripts must be used for Group A and Group B, Course Outcomes and Bloom's Levels are mentioned in additional Columns]

Course Outcomes (COs) of the Questions	
CO1	Demonstrate the understanding of the basic principles and operations on set theory, mathematical operations with complex numbers, geometrical interpretation and the related fundamental theories involving complex functions and the concept of transformation in a complex plane and waves
CO2	Apply the concept of transformation of an object into complex space and different mathematical operation of complex functions
CO3	Use Fourier series, Laplace's Transforms, Fourier Transform in different scenario
CO4	Analyze the harmonics & spectrum of different types of waves
CO5	Demonstrate the harmonic analysis using MATLAB

Bloom's Levels of the Questions						
Letter Symbols	R	U	App	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Group A

Answer the following questions

1. Define Fourier series in the interval $(-L, L)$. Sketch the following function for four cycles, CO3 U

$$y = f(t) = \begin{cases} 0; & -4 \leq t < 0 \\ 4; & 0 \leq t < 4 \end{cases}$$

Also find the Fourier series for the function.

OR

Derive the complex form of Fourier series.

2. a) Find Harmonic analysis of the given Fourier series CO4 An

$$f(t) = \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n+2}}{n} \sin \frac{n\pi}{3}$$

- b) Plot the line (at least 6) spectrum (discrete frequency spectra) for the Fourier series CO4 An

$$\underbrace{f(t)}_{\text{Complex wave}} = \underbrace{5}_{\text{DC value}} + \underbrace{\sum_{n=1}^{\infty} \frac{4}{n} \sin 2n\pi}_{\text{AC value}}$$

Group B
Answer the following questions

- 3 a) Prove that $L(f''(t)) = s^2 L\{f(t)\} - s f(0) - f'(0)$ CO3 Ap 5
 OR
 Find the inverse Laplace transform of $\frac{s+4}{s(s-1)(s-2)}$ U
- b) Express the following function in terms of unit step functions and find its Laplace transform CO3 Ap 5
 $f(t) = \begin{cases} 10; & t < 3 \\ 8; & t > 3 \end{cases}$
 OR
 Find Fourier Transform of
 $f(t) = \begin{cases} 1 & ; 0 \leq t < 1 \\ -1 & ; -1 \leq t < 0 \\ 0 & ; |t| > 1 \end{cases}$
- 4 a) State first shift theorem. Using the theorem evaluate $\mathcal{L}\{e^{-4t}t^2\}$ CO1 U
 b) Define unit step function. Sketch the following function, CO1 U
 $x(t) = -u(t+3) + 2u(t+1) - 2u(t-1) + u(t-3)$
- 5 a) Write **MATLAB code** to sketch line spectrum (at least 6) for the following Fourier series CO5 Ap

$$\underbrace{f(t)}_{\text{Complex wave}} = \underbrace{2}_{\text{DC value}} + \underbrace{\left[\sum_{n=1}^{\infty} (\cos n\pi + 1) \sin \frac{n\pi t}{3} \right]}_{\text{AC value}}$$
- b) Make a function in MATLAB environment to raise a complex wave $f(t)$ in the time interval of $[-4, 20]$ for the following Fourier series: CO5 Ap

$$f(t) = 4\pi + \sum_{n=1}^{\infty} \frac{3}{n\pi} \cos n\pi t$$
- c) If CO5 Ap
 $x[n] = \begin{cases} 5 & ; n = 0 \\ 6 & ; n = 1 \end{cases}$ and $h[n] = \begin{cases} 3 & ; n = 0 \\ -2 & ; n = 1 \end{cases}$
 Write MATLAB code to find the convolution sum of the above signals.

Time: 2 Hours 30 Minutes

Full Marks: 50

(i) The figures in the right-hand margin indicate full marks

(ii) Course Outcomes and Bloom's Levels are mentioned in additional Columns

Letter Symbols Meaning	Bloom's Levels of the Questions					
	R	U	App	An	E	C
	Remember	Understand	Apply	Analyze	Evaluate	Create

Group A

Answer the following questions

1. Define Fourier series in the interval $(-L, L)$. Sketch the following function for four cycles. CO3 App 10

$$y = f(t) = \begin{cases} 0; & -4 \leq t < 0 \\ 4; & 0 \leq t < 4 \end{cases}$$

Also find the Fourier series for the function

2. a) Find the Harmonic Analysis (at least 4) for the given Fourier series. CO4 An 5

$$f(t) = \frac{5}{2} - \frac{5}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} (\cos nx - 1) \sin \frac{n\pi t}{4}$$

Or

2. a) Plot the line (at least 6) spectrum (discrete frequency spectra) for the Fourier series CO4 An 5

$$\underline{f(t)} = \underline{\frac{5}{2}} + 2 \sum_{n=1}^{\infty} \frac{1}{n} \sin nt$$

2. b) Find Convolution sum $x[n] * h[n]$ for the following functions CO4 An 5

$$x[n] = 1; \quad n=0$$

$$= -1; \quad n=1$$

and

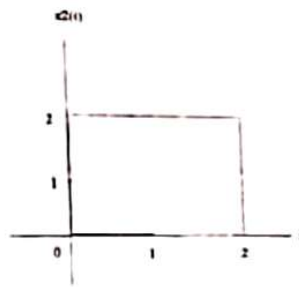
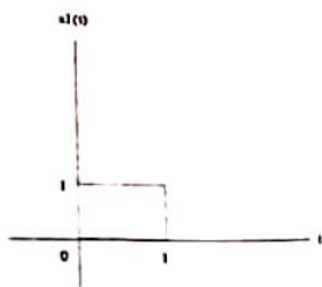
$$h[n] = 1; \quad n=0$$

$$= -1; \quad n=1$$

Or

Where n is the time index

2. b) Find convolution integral of $x(t) * x_2(t)$ CO4 An 5



Group B
Answer the following questions

~~5+4=2~~

3. Find the Fourier transform of the function $f(t) = \begin{cases} 1, & 0 < t < 1 \\ -1, & -1 < t < 0 \\ 0, & \text{otherwise} \end{cases}$

CO3 App 5

- (b) Find the inverse Laplace transform of $\frac{s+4}{s(s-1)(s-2)}$

CO3 App 5

Or:

- b) Find $L[u(t-3)] = \frac{e^{-3s}}{s}$

CO3 App 5

4. a) Solve the Initial Value Problem (IVP) by Laplace Transformation.

CO3 App 5

$$Y'' - 3Y' + 2Y = 4e^{2t}, \quad Y(0) = -3, \quad Y'(0) = 5$$

Or

- ✓ a) Solve the following Initial Value Problem (IVP) by Laplace Transform:

CO3 App 5

$$Y'' + Y = t, \quad Y(0) = 1, \quad Y'(0) = -2$$

- ✓ b) Define unit step function. Sketch the waveforms for the following signals

CO1 App 5

$$x(t) = -u(t+3) + 2u(t+1) - 2u(t-1) + u(t-3)$$

5. Write MATLAB code to sketch line spectrum (at least 6) for the following Fourier series

CO5 App 5

$$f(t) = \underbrace{2.5}_{\text{Complex value}} - \underbrace{\left[-\frac{5}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} (\cos n\pi - 1) \sin \frac{n\pi t}{4} \right]}_{\text{AC value}}$$

- b) Make a function in MATLAB environment to raise a complex wave $f(t)$ in the time interval of $[-4, 20]$ for the following Fourier series:

CO5 App

$$f(t) = \pi - \sum_{n=1}^{\infty} \frac{2}{n} \sin n\pi$$

- ✗ If $a[n] = \begin{cases} 2; & n=0 \\ -1; & n=1 \end{cases}$ and $b[n] = \begin{cases} 2; & n=0 \\ -2; & n=1 \end{cases}$ Write MATLAB code to find the convolution sum of the signals.

Page 2